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Cerebral near-infrared spectroscopy monitoring for prevention of death or neurodevelopmental disability in very preterm infants (Review)

Bodrero E, Isaza-López MC, Fiander M, Greisen G, Gluud C, Bruschettini M, supported by Cochrane Sweden and Cochrane Neonatal

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[Intervention Review]

Cerebral near-infrared spectroscopy monitoring for prevention of death or neurodevelopmental disability in very preterm infants

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ABSTRACT

Rationale

Very preterm infants (i.e. born before 32 weeks of gestation) are at risk of cerebral injury and long-term neurodevelopmental impairment. Cerebral near-infrared spectroscopy (NIRS) enables continuous monitoring of cerebral oxygenation to guide clinical management. Interest in NIRS has grown in recent years, highlighting the need for better evidence to support its clinical efficacy in improving brain development and reducing neurological sequelae. This is an update of a Cochrane review first published in 2017.

Objectives

To evaluate the beneficial and harmful effects of cerebral near-infrared spectroscopy (NIRS) monitoring versus no NIRS or blinded NIRS monitoring in very preterm infants.

Search methods

We searched CENTRAL, MEDLINE, Embase, CINAHL, three trial registries, and conference abstracts up to August 2025. We also checked reference lists of included studies and relevant systematic reviews.

Eligibility criteria

We included randomised clinical trials (RCTs) comparing cerebral NIRS monitoring versus no NIRS or blinded NIRS monitoring (where the treating healthcare professionals were unaware of the oxygenation levels) in very preterm infants.

Outcomes

Our critical outcomes included all-cause mortality at longest follow-up, major neurodevelopmental disability in children aged 18 to 24 months (a composite outcome including cerebral palsy, severe neurodevelopmental impairment, blindness, and profound hearing impairment), and major brain injury prior to discharge.

Our important outcomes included chronic lung disease at 36 weeks' gestational age, proven necrotising enterocolitis prior to discharge, retinopathy of prematurity (stage $\geq$ III) prior to discharge, and severe adverse reactions prior to discharge.

Risk of bias

We used Cochrane's original risk of bias tool (RoB 1).

Synthesis methods

We conducted meta-analyses using fixed-effect models to calculate risk ratios (RRs) and 95% confidence intervals (CIs) for all outcomes. We summarised the certainty of the evidence according to GRADE methods.

Included studies

We included five parallel-group RCTs published between 2016 and 2023 that enrolled a total of 2415 infants, with sample sizes ranging from 23 to 1600 infants per trial. The mean gestational age ranged from 26.1 weeks to 33.1 weeks. Four trials were conducted in multiple hospitals in high-income countries across North America, Asia, and Europe. The remaining trial took place in a single hospital in Austria. The comparator was no NIRS in two trials and blinded NIRS in three trials. In all trials, infants in the NIRS group were treated according to brain oxygen saturation values with specific preset treatment regimens, while those in the control group received standard or usual care regardless of NIRS monitoring values. The trials involved infants with varying start times and durations of NIRS monitoring. Only one trial reported major neurodevelopmental disability, and two trials reported retinopathy of prematurity (stage $\geq$ III). All five trials provided data for the other main outcomes of this review. We identified five ongoing trials.

Synthesis of results

NIRS monitoring compared with no NIRS or blinded NIRS monitoring likely results in little to no difference in all-cause mortality at longest follow-up (RR 0.99, 95% CI 0.82 to 1.18; $I^2 = 46\%$; 5 studies, 2415 participants; moderate-certainty evidence) and major brain injury diagnosed by brain ultrasound prior to discharge (RR 0.99, 95% CI 0.84 to 1.17; $I^2 = 13\%$; 5 studies, 2415 participants; moderate-certainty evidence). The evidence is very uncertain about the effect of NIRS monitoring compared with blinded NIRS monitoring on major neurodevelopmental disability in children aged 18 to 24 months (RR 1.28, 95% CI 0.50 to 3.29; 1 study, 115 participants; very low-certainty evidence).

NIRS monitoring compared with no NIRS or blinded NIRS monitoring likely results in little to no difference in chronic lung disease at 36 weeks of gestational age (RR 0.95, 95% CI 0.86 to 1.06; $I^2 = 43\%$; 5 studies, 2415 participants; moderate-certainty evidence), proven necrotising enterocolitis prior to discharge (RR 1.08, 95% CI 0.85 to 1.37; $I^2 = 0\%$; 5 studies, 2415 participants; moderate-certainty evidence), retinopathy of prematurity (stage $\geq$ 3) prior to discharge (RR 1.15, 95% CI 0.86 to 1.54; $I^2 = 0\%$; 2 studies, 1745 participants; moderate-certainty evidence), and severe adverse reactions prior to discharge (RR 9.41, 95% CI 0.51 to 174.44; I^2 not applicable; 5 studies, 2415 participants; moderate-certainty evidence).

Authors' conclusions

Overall, cerebral NIRS monitoring in very preterm infants likely results in little to no benefit for most measured outcomes.

Compared with conventional monitoring, cerebral NIRS monitoring in very preterm infants likely results in little to no difference in all-cause mortality at longest follow-up and in major brain injury diagnosed by brain ultrasound prior to discharge, and we are unsure about its effect on major neurodevelopmental disability in children aged 18 to 24 months. Furthermore, NIRS monitoring likely results in little to no difference in the risk of chronic lung disease at 36 weeks' gestational age, proven necrotising enterocolitis prior to discharge, severe retinopathy of prematurity prior to discharge, and severe adverse reactions prior to discharge.

Future randomised trials in very preterm infants should provide continuous NIRS monitoring from birth until cardiorespiratory stability to more accurately assess the potential benefits of the intervention. Further research is needed to understand and quantify performance differences among available NIRS devices and to evaluate their effects on long-term clinical outcomes. Few (if any) vital sign monitoring methods have demonstrated patient-relevant benefits in RCTs. For cerebral oximetry in preterm infants, trials using clinically meaningful endpoints (e.g. neurodevelopment at two years assessed with the Bayley Scales) may be infeasible due to the large sample sizes required. In this context, surrogate outcomes, such as electrophysiological markers of hypoxic brain injury, may offer a feasible alternative, provided they are rigorously validated against clinical endpoints.

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Registration

Protocol (2015): doi.org/10.1002/14651858.CD011506

Original review (2017): doi.org/10.1002/14651858.CD011506.pub2

Review update (2025): doi.org/10.1002/14651858.CD011506.pub3

PLAIN LANGUAGE SUMMARY

Does non-invasive monitoring (near-infrared spectroscopy) of oxygen in brain tissue help prevent brain injury in very premature babies?

Key Messages

- The evidence indicates that non-invasive monitoring (near-infrared spectroscopy; NIRS) of oxygen in brain tissue likely does not reduce death rates or short-term brain problems for very premature babies.
- Future studies should use NIRS more consistently and look at long-term effects to get clearer answers about potential benefits.

What is near-infrared spectroscopy (NIRS)?

Near-infrared spectroscopy (NIRS) is a non-invasive tool that can be used to measure brain oxygen levels.

How might NIRS help preterm babies in the neonatal intensive care unit?

Premature babies, especially those born very early (before 32 weeks of pregnancy), often face serious health problems. These potential problems include brain damage caused by insufficient oxygen supply to the brain. Using NIRS to monitor brain oxygen levels might help doctors take action early and quickly if something goes wrong, but it is important to know whether it actually works.

What did we want to find out?

We wanted to know whether NIRS monitoring is better than regular care for protecting the brain, improving neurological (brain) development, and helping very premature babies survive better. Specifically, we were interested in the effect of NIRS on:

- death for any reason;
- neurodevelopmental problems (including cerebral palsy, serious developmental delays, blindness, and deafness) when the babies grow older (18 to 24 months);
- serious brain damage;
- chronic lung disease;
- necrotising enterocolitis (serious intestinal disease);
- severe retinopathy of prematurity (eye disease); and
- severe unwanted reactions.

What did we do?

We searched for studies that compared NIRS monitoring with no NIRS or blinded NIRS monitoring (where the treating health professionals are unaware of the brain oxygen levels so cannot make any decisions based on this information) in very preterm infants. We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

We included five studies published between 2016 and 2023 and involving 2415 very preterm babies (1191 in the NIRS group and 1224 in the no NIRS/blinded NIRS group). Babies in the no NIRS/blinded NIRS group received standard care.

What are the main results?

Evidence from five studies (with 2415 infants) shows that NIRS monitoring compared with usual care likely results in little to no difference in:

- death for any reason;
- serious brain damage;
- chronic lung disease;
- necrotising enterocolitis (serious intestinal disease); and
- severe adverse reactions.

The evidence is very uncertain about the effect of NIRS on long-term neurodevelopmental problems when babies are older, based on one study with 115 babies. NIRS monitoring likely results in little to no difference in reducing severe retinopathy of prematurity (eye disease), based on two studies with 1745 babies.

What are the limitations of the evidence?

Four of the five included studies were small, and few infants died or had serious brain injury or long-term neurodevelopmental problems. In addition, the results varied widely across the different studies. Not all studies used NIRS in the same way, which makes it hard to know

if it would have the same effects for all babies across the different health centres in different countries. Finally, not all the outcomes we were interested in were clearly measured.

The results of future research could differ from the results of this review.

How up to date is this evidence?

The evidence is current to August 2025. Five relevant studies are in progress, and one of these plans to enrol 3000 babies by 2029. This large study may provide more evidence to inform our understanding of the benefits and harms of NIRS in very preterm infants.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Continuous cerebral NIRS monitoring compared to conventional monitoring with no NIRS or blinded NIRS monitoring in very preterm infants (gestational age < 32 weeks)

Continuous cerebral NIRS monitoring compared to conventional monitoring with no NIRS or blinded NIRS monitoring in very preterm infants (gestational age < 32 weeks)

Patient or population: very preterm infants (gestational age < 32 weeks)

Setting: neonatal units (mostly in high-income countries) in Asia, Europe, and North America

Intervention: continuous cerebral NIRS monitoring

Comparison: conventional monitoring with no NIRS or blinded NIRS monitoring

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with conventional monitoring with no NIRS or blinded NIRS monitoring	Risk with continuous cerebral NIRS monitoring				
All-cause mortality at longest follow-up	163 per 1000	162 per 1000 (193 to 134)	RR 0.99 (0.82 to 1.18)	2415 (5 RCTs)	⊕⊕⊕⊖ Moderate ^a	NIRS likely results in little to no difference in all-cause mortality at longest follow-up.
Major neurodevelopmental disability (cerebral palsy, severe neurodevelopmental impairment, blindness, or profound hearing impairment) in children aged 18–24 months	120 per 1000	154 per 1000 (395 to 60)	RR 1.28 (0.50 to 3.29)	115 (1 RCT)	⊕⊖⊖⊖ Very low ^b	The evidence is very uncertain about the effect of continuous cerebral NIRS monitoring on major neurodevelopmental disability in children aged 18–24 months.
Major brain injury (intra-ventricular haemorrhage > grade III or periventricular cystic leukomalacia) diagnosed by brain ultrasound prior to discharge	185 per 1000	183 per 1000 (216 to 155)	RR 0.99 (0.84 to 1.17)	2415 (5 RCTs)	⊕⊕⊕⊖ Moderate ^a	Continuous cerebral NIRS monitoring likely results in little to no difference in major brain injury diagnosed by brain ultrasound prior to discharge.
Chronic lung disease at 36 weeks of gestational age	372 per 1000	353 per 1000 (394 to 320)	RR 0.95 (0.86 to 1.06)	2415 (5 RCTs)	⊕⊕⊕⊖ Moderate ^a	Continuous cerebral NIRS monitoring likely results in little to no difference in chronic lung disease at 36 weeks of gestational age.

Proven necrotising enterocolitis (Bell's stage II or III) prior to discharge	95 per 1000	102 per 1000 (130 to 81)	RR 1.08 (0.85 to 1.37)	2415 (5 RCTs)	⊕⊕⊕⊖ Moderate ^a	Continuous cerebral NIRS monitoring likely results in little to no difference in proven necrotising enterocolitis prior to discharge.
Retinopathy of prematurity (stage ≥ 3) prior to discharge	87 per 1000	100 per 1000 (134 to 75)	RR 1.15 (0.86 to 1.54)	1745 (2 RCTs)	⊕⊕⊕⊖ Moderate ^a	Continuous cerebral NIRS monitoring likely results in little to no difference in retinopathy of prematurity (stage ≥ 3) prior to discharge.
Severe adverse reactions prior to discharge	0 per 1000	0 per 1000 (0 to 0)	RR 9.41 (0.51 to 174.44)	2415 (5 RCTs)	⊕⊕⊕⊖ Moderate ^c	Continuous cerebral NIRS monitoring likely results in little to no difference in severe adverse reactions prior to discharge.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_452475607216890995.

^a Downgraded one level for serious imprecision (wide CI). Not downgraded for overall high risk of bias due to the objective nature of the measured outcome.

^b Downgraded three levels for extremely serious imprecision (very wide CIs, small sample size, few events). Not downgraded for overall high risk of bias due to the objective nature of the measured outcome.

^c Downgraded one level for serious imprecision (very few events). Not downgraded for overall high risk of bias due to the objective nature of the measured outcome.

BACKGROUND

Description of the condition

Preterm birth carries a high risk of cerebral impairment and death. Twenty-five per cent of infants born more than 12 weeks before term die before hospital discharge, and about 25% of survivors will have cerebral palsy or other neurodevelopmental delays [1, 2, 3].

The aetiology behind the long-term complications of preterm birth is multifactorial. Studies indicate that cerebral hypoxia-ischaemia plays an important role [4, 5]. Hypoxia-ischaemia represents a mismatch between cerebral oxygen supply and demand that can be caused by hypotension [6], low systemic blood flow [7], increased cerebral oxygen demand, or low arterial blood oxygen content. Conversely, some research suggests that too much oxygen is deleterious to the immature brain [8].

Importantly, preterm brain injury is no longer viewed solely as a destructive process. Instead, it is now understood as a complex amalgam of both acute cellular injury and disrupted brain maturation, a concept Volpe and colleagues called "encephalopathy of prematurity" [3]. This framework highlights that preterm infants not only experience neuronal and glial cell death, but also impaired processes such as myelination, synaptogenesis, and cortical development.

Specific brain lesions thought to be associated with hypoxia-ischaemia include germinal matrix haemorrhage, intraventricular haemorrhage [9, 10], and periventricular leukomalacia [11, 12], all of which contribute to both destructive injury and long-term disruption of neurodevelopmental trajectories.

Moreover, while hyperoxia may harm the developing brain, it is also implicated in injury to other vulnerable tissues. Excess oxygen exposure is linked to retinopathy of prematurity (a leading cause of childhood blindness) and bronchopulmonary dysplasia, a chronic lung disease. Hyperoxia impairs retinal vascular development via vascular endothelial growth factor (VEGF)-mediated neovascularisation, and leads to alveolar and vascular lung injury [13]. Recent retrospective studies reinforce these connections, noting that non-invasive ventilation and supplemental oxygen use heighten the risk of retinopathy of prematurity in infants with bronchopulmonary dysplasia, especially those with low gestational age and birthweight [14].

To prevent or minimise permanent neurological damage from hypoxia-ischaemia or hyperoxia, immediate and appropriate interventions are essential, including therapeutic hypothermia, optimal mechanical ventilation strategies, and comprehensive supportive care such as fluid resuscitation, inotropic support, and blood transfusions to enhance cerebral oxygen delivery [15, 16, 17]. However, knowing when to implement these interventions remains a challenge, as cerebral hypoxia-ischaemia often has few clinical signs or symptoms [18].

Description of the intervention and how it might work

Near-infrared spectroscopy (NIRS) enables measurement of regional tissue haemoglobin oxygen saturation (rStO₂) [19]. rStO₂ represents a weighted average of arterial and venous oxygen saturation. The weight is commonly assumed to be 25% arterial and 75% venous, but may vary within and among patients [20]. There is currently no established reference standard. Several studies have

found a significant correlation between rStO₂ measured by cerebral NIRS and venous jugular bulb blood oxygen saturation, but the agreement is generally poor ($\pm 17\%$ limits of agreement) [21, 22], possibly because NIRS is a volume-weighted measure, whereas jugular bulb oxygen saturation is a flow-weighted measure.

The relationship between cerebral rStO₂ and brain damage has been assessed in several piglet studies. It seems that both the magnitude of cerebral rStO₂ deviation from normality and the duration of deviation are important factors [23, 24, 25]. It is thus reasonable to assume that rStO₂ must be kept within a safe range and that brief or minor deviations from that range are harmless.

Data on the clinical significance of cerebral rStO₂ are still rather scant. In adults, Heringlake and colleagues found that a precardiac surgery cerebral rStO₂ level below 50% was an independent risk factor for 30-day and one-year mortality [26]. Furthermore, studies have associated intraoperative desaturation during cardiac surgery with postoperative cognitive dysfunction [27, 28]. In very preterm infants, Selvanathan and colleagues recently showed that early-life exposure to stressors such as invasive procedures significantly affects brain connectivity and neurodevelopment, underscoring the clinical relevance of continuous cerebral oxygen monitoring [2].

Toet and colleagues found that rStO₂ was a predictor of neurodevelopmental outcomes in term newborns with hypoxic-ischaemic encephalopathy [29]. Children with adverse outcomes had a significantly elevated cerebral rStO₂ at 24 hours postpartum compared to infants with normal outcomes. In preterm infants, Sorensen and colleagues observed a correlation between the degree of cerebral hypoxia estimated by NIRS and the severity of brain damage [30]. This was further supported by Thewissen and colleagues, who demonstrated that impaired cerebral autoregulation and prolonged cerebral hypoxia were associated with intraventricular haemorrhage or early death in hypotensive preterm neonates [5]. Two other studies in preterm infants found an association between low rStO₂ and cerebral injury [9, 31], while a systematic review by Rees and colleagues highlighted the impact of even low-grade intraventricular haemorrhage on long-term neurodevelopment [1]. In accordance with these findings, Alderlisten and colleagues found that having an rStO₂ below 50% for more than 10% of the time during the first three days of life was associated with a five-point lower developmental quotient [32].

The precise safe range of cerebral rStO₂ is unclear, and both inter- and intra-individual variance is likely. In adults undergoing surgery, thresholds for intervention may be dependent on baseline rStO₂ (e.g. intervene when rStO₂ has dropped 20%) [33]. For preterm infants, who can be haemodynamically unstable from birth, the usefulness of such a baseline measurement is less evident. Thresholds for intervention must rely on absolute values of rStO₂. This practice is analogous to the use of pulse oximetry to guide the adjustment of the fraction of inspired oxygen for newborn infants [34]. This is problematic because repeated measures within the same subject have a within-subject standard deviation (SD) of approximately 5% to 6% [30, 35], while there is a systematic bias between sensors from INVOS 5100 and NIRO 300 [36]. In addition, Hessel and colleagues found that the difference in rStO₂ estimates between INVOS 5100 and FORE-SIGHT increases with decreasing oxygenation in infants [37, 38].

Accumulating evidence suggests it is possible to minimise cerebral desaturation in preterm infants in the delivery room [39] and in the

first three days of life [40], and in adults undergoing cardiac surgery [41, 42] and non-cardiac surgery [43]. It remains to be established if this minimisation improves patient-relevant outcomes.

This review focused on cerebral NIRS in neonatal intensive care units (NICUs). The optimal timing of monitoring in preterm infants is unclear, but it should follow the notion that cerebral NIRS monitoring must be continuous to have the greatest possible impact on clinical outcomes. Moreover, as the first week of life is the most vulnerable time period with regard to low cerebral blood flow and brain damage [3, 44, 45], early NIRS monitoring within hours or a few days of birth is likely to be most efficient [46]. Most NIRS devices estimate rStO₂ every five seconds. The clinical interpretation is based not only on the current absolute cerebral rStO₂ value, but perhaps even more so on the trend in rStO₂ over minutes and hours.

NIRS-guided intensive care is a complex intervention, as it consists of monitoring a physiological quantity (similar to a diagnostic test) coupled with a number of possible specific interventions either given alone or in concert. In other words, NIRS monitoring will have no impact on clinical outcomes if it does not lead to changes in clinical management. In a trial setting, having prespecified treatment guidelines will increase the likelihood that clinicians deal with out-of-range cerebral rStO₂ values in a consistent manner. The appropriateness of the treatment guideline also depends on whether it defines cerebral rStO₂ thresholds for intervention, if these thresholds are appropriate, and if there is clear guidance on how to react outside these thresholds. For comparison, with pulse oximetry, a target range of 85% to 89% saturation has been associated with a lower survival rate compared with a target range of 91% to 95% in extremely preterm infants [47]. Cerebral rStO₂ is a more complex physiological measure than arterial oxygen saturation (SpO₂), and its safe ranges are likely to be broader. Furthermore, as described above, appropriate rStO₂ thresholds could well be device dependent. The overall intervention quality depends on the NIRS monitoring and the treatment guideline describing how to normalise rStO₂, as well as adherence to the guideline; differences in compliance could be a source of heterogeneity. Although meta-analyses of complex interventions should incorporate a measure of their quality [48], there is no generally accepted framework for such an analysis. Moreover, few trial reports provide sufficient detail on how to assess intervention quality [49]. We proposed subgroup analyses based on duration of NIRS monitoring, assessment of trial treatment guidelines, and adherence to these guidelines.

Adverse effects directly relating to the NIRS device are similar to those of the widely applied pulse oximetry, primarily local skin damage such as burns, pressure sores, and skin irritation [50]. Another concern is that NIRS-driven interventions could be deleterious. SpO₂ is related to rStO₂, and it is probably inappropriate to aim for SpO₂ higher than the normal target range to increase a low rStO₂. Such an intervention could increase the incidence of retinopathy and other oxygen damage.

By alerting healthcare professionals when cerebral hypoxia or hyperoxia occurs, cerebral NIRS monitoring could improve the clinical outcomes of preterm infants. Three variables control cerebral rStO₂: oxygen delivery to the brain, cerebral oxygen consumption, and the arterial-venous volume ratio. Cerebral oxygen consumption probably varies little in preterm infants, and the arterial-venous volume ratio is irrelevant for cerebral

oxygen sufficiency. Therefore, interventions guided by cerebral oxygenation in preterm infants focus on oxygen delivery to the brain, which depends on the oxygen content of arterial blood, cerebral blood flow, or the combination of both. Interventions that alter the oxygen content include changing fractional inspiratory oxygen or elevating haematocrit by blood transfusion, while clinicians can alter cerebral blood flow by changing mean airway pressure and ventilatory rate during mechanical ventilation or by increasing blood pressure and cardiac output by fluid, vasopressors, or inotropes. Observation of abnormal values could thus motivate reactive interventions such as changes in ventilator settings and initiation of haemodynamic support. Possible relevant interventions depend on the situation at hand [51]. The main goal is to respond to NIRS measurement of cerebral rStO₂ and initiate treatments known to normalise cerebral blood flow or blood oxygen content [33].

Why it is important to do this review

This is an update of a Cochrane review first published in 2017 [52]. We considered an update necessary because the previous review included only one trial and found insufficient evidence to identify the benefits and harms of cerebral NIRS oximetry in preterm infants. Moreover, there is a growing interest among clinicians in the use of NIRS in preterm infants to improve brain development and reduce neurological sequelae in the long term. Greater use of cerebral NIRS requires a better understanding of its clinical efficacy.

OBJECTIVES

To evaluate the beneficial and harmful effects of cerebral near-infrared spectroscopy (NIRS) monitoring versus no NIRS or blinded NIRS monitoring in very preterm infants.

METHODS

We conducted and reported this review following methods and standards described in the *Cochrane Handbook for Systematic Reviews of Interventions* [53], the *Methodological Expectations of Cochrane Intervention Reviews (MECIR) Manual* [54], and the PRISMA 2020 Statement [55, 56].

Changes made since the 2017 review

All decisions regarding changes to the previous review were made by consensus after a discussion between the review authors.

- The title of the original review and protocol was *Cerebral near-infrared spectroscopy monitoring for prevention of brain injury in very preterm infants* [52, 57]. We changed the title to better focus on critical outcomes 'death and neurodevelopmental disability', while 'brain injury' can be associated with a broad clinical spectrum from mild to severe neurological pictures.
- We use a study-centric format to managing data in our review.
- We retained the same critical outcomes as in the original review, but reduced the number important outcomes from 18 to eight to better focus the review and make it more user-friendly [58]. The outcomes we removed were:
 - number of changes in mean airway pressure per time-unit (i.e. to increase or decrease continuous positive airway pressure or positive end-expiratory pressure);
 - number of changes per time-unit in inspiratory oxygen fraction FiO₂;

- number of changes per time-unit in blood pCO₂ (i.e. to change ventilation by changing respiratory rate on mechanically ventilated infants);
- number of changes per time-unit in vasopressor or inotrope dosage, or both (i.e. to initiate, stop, increase, or decrease the use of vasopressors/inotropic agents);
- number of fluid bolus treatments (i.e. to give a bolus of fluid at 5 to 20 ml/kg bodyweight) to increase systemic blood circulation per time-unit;
- number of blood transfusions administered;
- number of patent ductus arteriosus treatments (i.e. attempt to close the duct medically or surgically) received;
- intraventricular haemorrhage (all grades);
- patent ductus arteriosus diagnosed by echocardiography;
- renal impairment (creatinine ≥ 120 $\mu\text{mol/L}$, oliguria ≤ 0.5 ml/kg/hour at 1 week of age); and
- adverse effects not listed as an outcome above but reported by the authors as an adverse event.
- We added 'abnormal fidgety movements' as an important outcome because it is an important early predictor of brain damage.
- We amended 'mortality' by changing the time points of measurement to 'prior to discharge, at two years of life, and at longest follow-up'. This was to obtain even more details on an outcome we considered crucial.
- We removed a minimal duration of NIRS monitoring to be more inclusive and avoid bias in the selection of trials.
- We selected a few different outcomes to include in the summary of finding tables and on which to perform a GRADE assessment, preferring to include all three critical outcomes and reducing the total number to seven. Specifically, we included 'major neurodevelopmental disability in children aged 18 to 24 months (composite outcome)', 'major brain injury diagnosed by brain ultrasound prior to discharge', and 'severe adverse reactions as reported by study authors prior to discharge' in place of 'intraventricular haemorrhage grade III or IV', 'cystic periventricular leukomalacia', 'patent ductus arteriosus', and 'other adverse effects'.
- We did not perform sensitivity analysis to handle missing data, because there were no missing data ([Dealing with missing data; Sensitivity analysis](#)).
- We removed the intervention quality subgroup analysis because we addressed this aspect with the risk of bias tool ([Risk of bias assessment in included studies](#)) and GRADE ([Certainty of the evidence assessment](#)).
- We agreed these changes with Cochrane Neonatal, following MECIR standard U1 ("It will be necessary to agree the approach to updating the review with the CRG").
- We moved content about trial sequential analysis (TSA) to the [Sensitivity analysis](#). We planned to carry out a sensitivity analysis of GRADE imprecision, with results reported in a separate section ([Results](#)).
- We did not address the issue of the different brands of NIRS devices used within a given trial.

Criteria for considering studies for this review

Types of studies

Randomised clinical trials (RCTs), cluster-RCTs, and quasi-RCTs (studies that used an approximation of randomisation to allocate participants to treatment) were eligible for inclusion.

Types of participants

Eligible participants were very preterm infants (born before 32 weeks of gestation) or very-low-birthweight (VLBW) infants (birthweight < 1500 g). We included both singletons and infants from multiple births (see [Unit of analysis issues](#)). We chose these cut-offs (32 weeks and 1500 g) based on previous studies that investigated the risk of brain injury associated with decreasing gestational age and birthweight [59, 60, 61]. If we identified studies with mixed populations, we included data if at least 50% of participants met our eligibility criteria.

Types of interventions

Our comparison of interest was continuous cerebral NIRS monitoring versus conventional monitoring with no NIRS, or blinded NIRS monitoring during intensive care. The monitoring had to start within one week from birth and be continuous. We applied no limitations related to the duration of NIRS monitoring. We included trials where detection of rStO₂ values led to changes in treatment in the experimental group. The only prerequisite for the NIRS equipment was that it had to show a value for tissue oxygenation as a percentage.

Possible controls (conventional monitoring) included invasive or non-invasive blood pressure monitoring, pulse oximetry, electrocardiographic monitoring, transcutaneous oxygen partial pressure (pO₂) and carbon dioxide partial pressure (pCO₂) monitoring, or any combination of these, without NIRS monitoring or with blinded NIRS monitoring (where the treating healthcare professionals were unaware of the rStO₂ values).

Concomitant interventions or treatments were allowed, provided the intervention groups were offered these treatments equally.

We planned to exclude trials that used NIRS monitoring as part of multi-monitoring (i.e. where monitoring modalities other than NIRS differed between groups).

Outcome measures

We included studies that planned to measure the following outcomes, irrespective of whether they reported the relevant data.

Critical outcomes

- All-cause mortality prior to discharge, at two years of life, and at longest follow-up
- Major neurodevelopmental disability in children aged 18 to 24 months (composite outcome including cerebral palsy, severe neurodevelopmental impairment as evaluated by a validated developmental assessment tool (e.g. more than 2 SDs below the mean in Bayley Scales of Infant Development), blindness (vision < 6/60 in both eyes), and profound hearing impairment). We planned to report these components of major neurodevelopmental disability for all trials that evaluated children after 18 months' chronological age, and to perform

separate analyses for children aged 18 to 24 months (primary outcome) and three to five years.

- Major brain injury, defined as intraventricular haemorrhage greater than grade III or periventricular cystic leukomalacia [62], diagnosed by brain ultrasound or magnetic resonance imaging (MRI) prior to discharge [63, 64]

Important outcomes

- Intraventricular haemorrhage grade III or IV prior to discharge [65]
- Cystic periventricular leukomalacia prior to discharge
- Air leak (any and gross, including pneumothorax or pneumomediastinum) prior to discharge
- Chronic lung disease (oxygen supplementation) at 36 weeks of gestational age
- Proven necrotising enterocolitis prior to discharge (Bell's stage II or III) [66]
- Retinopathy of prematurity (any stage) prior to discharge
- Severe retinopathy of prematurity (stage $\geq$ III) prior to discharge [67]
- Severe adverse reactions as reported by study authors prior to discharge
- Abnormal fidgety movements at six to 20 weeks of corrected age

Search methods for identification of studies

Electronic searches

One review author (MF) revised our search strategies for this update to increase transparency (documenting strategies exactly as run) and sensitivity. We ran the searches with no language or publication status restrictions, but we did use methodological filters to limit the results (sources of filters provided in search strategies). Searches for systematic reviews were limited to the past two years. Searches for trials had no date limits, so we deduplicated them against results screened for the 2017 version of this review. We searched the following databases on 15 August 2025.

- Ovid MEDLINE All (1946–15 August 2025).
- Ovid Embase (1974–15 August 2025).
- Cochrane Register of Studies.
- CINAHL Complete, EVSCO (1981–15 August 2025).

We searched the following three trials registries.

- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (<https://clinicaltrials.gov/>).
- World Health Organization International Clinical Trials Registry Platform (<https://trialsearch.who.int/>).
- ISRCTN Registry (<https://www.isrctn.com/>).

All search strategies are provided in [Supplementary material 1](#).

Searching other resources

We conducted manual searches of the following conference abstracts.

- European Academy of Paediatric Societies (EAPS; 2020–2024).
- Pediatric Academic Societies (PAS; 2019–2020, 2023–2025).
- Perinatal Society of Australia and New Zealand (PSANZ; 2020–2024.)

We searched for retractions or errata notices for our included studies in the Retraction Watch Database and PubMed on 21 August 2025.

We asked experts in the NIRS field about their knowledge of unpublished RCTs.

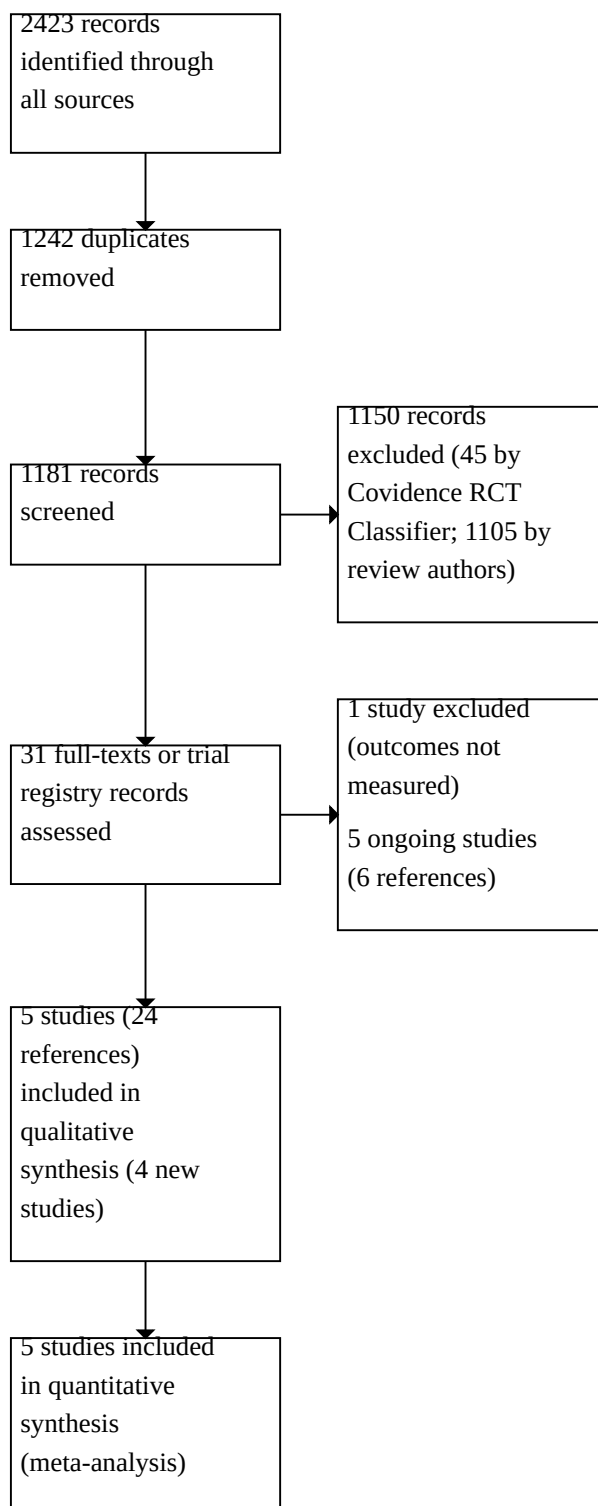
Data collection and analysis

Where we identified studies involving authors of this review, other authors independently made decisions about inclusion, extracted data, assessed risk of bias, and assessed the certainty of the evidence.

Selection of studies

We screened the search results using Covidence [68]. Two review authors (from EB, MCI, or MB) independently screened the titles and abstracts, then assessed the full-text articles of potentially eligible records. Disagreements arising at any point in the screening process were resolved by discussion with the entire review author team. We collated multiple reports of the same study under a single study ID so that each study, rather than each report or reference, was the unit of interest. [Figure 1](#) illustrates the selection process in a PRISMA flow diagram [55, 56]. We documented information about studies excluded during full-text screening and reasons for exclusion.

Figure 1. PRISMA flow diagram illustrating the study selection process



Data extraction and management

Using Covidence [68], two review authors (EB, MCI) independently extracted data to a form that we had piloted on two studies. Disagreements were resolved through discussion, or by input from another review author (MB). We contacted the authors of two trials to request additional information (Pichler 2016 [69, 70]; Pichler 2018 [71, 72]).

We extracted the following characteristics from each included study.

- Administrative details: study author(s), published or unpublished, year of publication, year in which study was conducted, presence of vested interest, funding.
- Study characteristics: study registration or protocol (if available), study design, study setting, number of study centres and location(s), informed consent, ethics approval, completeness of follow-up (e.g. greater than 80%).
- Participants: number randomised, number lost to follow-up/withdrawn, number analysed, mean gestational age, gestational age range, mean chronological age, chronological age range, inclusion and exclusion criteria.
- Interventions.
- Outcomes as described in [Outcome measures](#).

After extracting all data of interest, we imported them from Covidence into RevMan [68, 73]. We replaced any standard error of the mean (SEM) with the corresponding SD.

We recorded available information about ongoing studies, including trial completion date. We listed no studies as awaiting assessment.

Risk of bias assessment in included studies

To assess risk of bias, we used the original Cochrane risk of bias tool (RoB 1), which covers the following domains [74].

- Sequence generation (selection bias).
- Allocation concealment (selection bias).
- Blinding of participants and personnel (performance bias).
- Blinding of outcome assessment (detection bias).
- Incomplete outcome data (attrition bias).
- Selective reporting (reporting bias).
- Any other bias.

Two review authors (EB, MCI) independently rated trials at low, unclear, or high risk of bias in each domain, resolving disagreements by discussion or in consultation with a third review author (MB). They then assigned an overall risk of bias rating as follows.

- Low overall risk of bias: all domains classified at low risk of bias.
- Unclear overall risk of bias: one or more domains classified as unclear, and no domain at high risk of bias.
- High overall risk of bias: at least one domain classified at high risk of bias.

Measures of treatment effect

We present dichotomous outcomes as risk ratios (RRs) with naive 95% confidence intervals (CIs), with values less than 1.00 favouring

NIRS-guided treatment. We planned to calculate risk differences (RDs) with 95% CIs and convert them to number needed to treat for an additional beneficial/harmful outcome if there were statistically significant intervention effects.

Unit of analysis issues

The unit of analysis was the participating infant in individually randomised trials, and each infant was considered only once in each analysis. We found no cluster-RCTs, cross-over trials, or multiarm trials.

Dealing with missing data

We planned to contact the original investigators and study authors for any missing data then use a 'complete-participant analysis', excluding all participants with missing outcome data (see [Sensitivity analysis](#) section). However, no outcome data were missing.

Reporting bias assessment

We would have assessed reporting biases using funnel plots of the effect estimates if any meta-analyses had included 10 or more trials.

Synthesis methods

We used a fixed-effect model to combine data where it was reasonable to assume that studies were estimating the same underlying treatment effect. Cochrane neonatal reviews typically use a fixed-effect model, for two main reasons: preterm neonates are relatively similar in terms of their general condition, as they are less likely to be influenced by confounding factors that take time to develop; and the inclusion criteria reflect narrow research questions. In addition, because the NICU is a controlled environment with standard basic care protocols, interventions administered in this setting tend to be relatively easy to standardise. Taking this into consideration, a fixed-effect model is more sensitive in detecting small effect sizes.

In assessing the treatment effects for dichotomous outcomes, we planned to use the RR and RD with naive 95% CIs. For outcomes measured on a continuous scale, we planned to use MDs with naive 95% CIs.

We tested for statistical heterogeneity using the Chi² test with significance set at $P < 0.1$, and we quantified inconsistency using the I^2 statistic, which describes the percentage of the variability in effect estimates that is due to heterogeneity rather than sampling error (chance). We interpreted the I^2 values as follows.

- Less than 25%: no heterogeneity.
- 25% to 49%: low heterogeneity.
- 50% to 74%: moderate heterogeneity.
- 75% and above: high heterogeneity.

We intended to explore the possible causes of statistical heterogeneity using planned and post-hoc subgroup analyses.

Investigation of heterogeneity and subgroup analysis

In subgroup analyses, we planned to examine the effectiveness and safety of NIRS-guided treatment in relation to the following criteria.

- Gestational age, possibly extracted from subgroups within each trial (≤ 28 weeks versus > 28 weeks).

- Duration of NIRS monitoring: the trials were to be ranked from shortest to longest NIRS duration and the subgroups defined as below the median or equal to and above the median.

We planned to conduct a test of interaction for the subgroups according to Altman and Bland [75].

Equity-related assessment

We did not address equity due to the nature of the intervention, likely to be used mainly in high-income countries.

Sensitivity analysis

For missing data, we planned to conduct sensitivity analyses for our primary outcomes by applying best- and worst-case scenarios. The best-case scenario was that all participants lost to follow-up in the experimental group had a beneficial outcome, and all participants lost to follow-up in the control group had a detrimental outcome. The worst-case scenario was that all participants lost to follow-up in the experimental group had a detrimental outcome, and all participants lost to follow-up in the control group had a beneficial outcome.

Trial Sequential Analysis

We presented dichotomous outcomes as RRs with Trial Sequential Analysis (TSA)-adjusted CIs [76, 77]. Our TSAs are presented in <https://doi.org/10.6084/m9.figshare.29948006.v1>. TSA figures were created using Copenhagen Trial Unit's TSA Software [78].

For dichotomous outcomes, we conducted TSA to calculate the required quantity of information (number of participants) and the cumulative Z-curve's breach of relevant trial sequential monitoring boundaries [79]. We estimated the diversity-adjusted required information size (DARIS) based on the proportion of participants with an outcome in the control group, a relative risk reduction (RRR) of 20%, an alpha of 5%, a beta of 10%, and the diversity in the meta-analysis. We used TSA to assess the need for any downgrading for imprecision and compare that to the downgrading for imprecision according to GRADE [80]. In TSA, we downgraded by three levels for imprecision if the accrued number of participants was below 33% of DARIS, two levels if between 34% and 66% of DARIS, and one level if between 67% and 99% of DARIS. We did not downgrade if the cumulative Z-curve crossed the monitoring boundaries for benefit, harm, or futility, or if DARIS was reached.

Certainty of the evidence assessment

Using GRADEpro GDT [81], we prepared a summary of findings table for one comparison: continuous cerebral NIRS monitoring versus conventional monitoring with no NIRS or blinded NIRS monitoring.

We used the GRADE approach, as outlined in the *GRADE Handbook*, to assess the certainty of the evidence for the following (clinically relevant) outcomes [80].

- All-cause mortality at longest follow-up.
- Major neurodevelopmental disability in children aged 18 to 24 months (composite outcome).
- Major brain injury diagnosed by brain ultrasound prior to discharge.
- Chronic lung disease (oxygen supplementation) at 36 weeks of gestational age.
- Proven necrotising enterocolitis prior to discharge.

- Retinopathy of prematurity (stage $\geq$ III) prior to discharge.
- Severe adverse reactions as reported by study authors prior to discharge.

Two review authors (EB, MCI) independently assessed the certainty of the evidence for each of the outcomes above. We considered evidence from RCTs as high certainty to begin with but downgraded for limitations based on the following five considerations.

- Design (risk of bias).
- Consistency across studies.
- Directness of the evidence.
- Precision of estimates.
- Presence of publication bias.

We interpreted the final GRADE ratings as follows.

- High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
- Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
- Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

Consumer involvement

No consumers were involved in this review, although we did base the review outcomes on core outcome sets developed with consumer involvement [58].

RESULTS

Description of studies

See [Supplementary material 2](#), [Supplementary material 3](#), and [Supplementary material 4](#).

Results of the search

Our searches identified 2423 references; after deduplication, 1181 were available for screening. After excluding 1150 records based on their title/abstract (45 through the Covidence RCT Classifier and 1105 through manual screening), we reviewed 31 full-text articles or trial registry records. We excluded one study, identified five ongoing studies (six references), and included five studies (24 references).

Four of the included studies are new to this update (Hansen 2023 [82, 83, 84, 85, 86, 87, 88, 89]; Pichler 2016; Pichler 2018; Pichler 2023 [90, 91, 92, 93, 94]). We included one previously excluded study (Pichler 2016), because we removed the time limitation for the intervention so that we could assess the effect of NIRS during immediate transition after birth.

The supplementary material contains characteristics of included studies ([Supplementary material 2](#)), excluded studies ([Supplementary material 3](#)), and ongoing studies ([Supplementary material 4](#)). [Figure 1](#) illustrates the study selection process in a PRISMA diagram.

Included studies

Five studies met our eligibility criteria (Hansen 2023; Hyttel-Sorensen 2015 [95, 96, 97, 98, 99, 100, 101]; Pichler 2018; Pichler 2023; Pichler 2016). [Table 1](#) presents an overview of the included studies, and [Supplementary material 2](#) provides more detailed characteristics.

Settings

Hansen 2023 was conducted across 17 countries in Asia, Europe, and North America. Hyttel-Sorensen 2015 was conducted in France, Denmark, Spain, Ireland, the Netherlands, Austria, Italy, and the UK. Pichler 2023 took place in seven European countries and Canada. Pichler 2018 was a single-centre study conducted in Austria. Pichler 2016 took place in Austria and Canada.

Participants

The five trials included 2580 infants, with sample sizes ranging from 60 infants (Pichler 2016) to 1601 infants (Hansen 2023). The mean gestational age was 26.1 weeks in Hansen 2023, 26.6 weeks in Hyttel-Sorensen 2015, 28.9 weeks in Pichler 2023, 29.2 weeks in Pichler 2016, and 33.1 weeks in Pichler 2018. Eligible gestational age at birth was less than 28 weeks in two trials (Hansen 2023; Hyttel-Sorensen 2015), less than 32 weeks in Pichler 2023, less than 34 weeks in Pichler 2016, and less than 37 weeks in Pichler 2018. Three trials excluded participants with congenital malformations (Pichler 2016; Pichler 2018; Pichler 2023).

Interventions

Two trials compared continuous NIRS monitoring to no NIRS (Hansen 2023; Pichler 2016). Three trials compared continuous NIRS monitoring to blinded NIRS (Hyttel-Sorensen 2015; Pichler 2018; Pichler 2023). Two trials started NIRS monitoring in the first six hours after birth (Hansen 2023; Pichler 2018), one trial in the first three hours after birth (Hyttel-Sorensen 2015), and two trials in the first two or three minutes after birth with a duration of 15 minutes (Pichler 2016; Pichler 2023). The mean duration of NIRS monitoring was 67.9 hours in Hyttel-Sorensen 2015 and 24.4 hours in Pichler 2018. Hansen 2023 planned 72 hours of NIRS monitoring, and only 4.7% of infants in the experimental group (36/772) had more than 14 hours of missing NIRS monitoring data (study authors' definition of nonadherence).

In all trials, infants in the NIRS group were treated according to brain oxygen saturation values with a prespecified treatment regimen. In the control group, infants received standard or usual care regardless of the NIRS monitoring values (where recorded).

The most commonly used NIRS monitoring device was the INVOS 5100 Cerebral/Somatic Oximeter (all NICUs in Pichler 2016 and Pichler 2023, and most NICUs in Hyttel-Sorensen 2015 and Hansen 2023). Additionally, all NICUs in Pichler 2018 and two NICUs in Hyttel-Sorensen 2015 used the NIRO 200NX. In Hansen 2023, some NICUs also used other oximetry devices approved for clinical use in neonates, such as OxyPrem, EGOS, and O3 neonatal sensors.

Outcomes

All five trials reported all-cause mortality prior to discharge and at longest follow-up, major brain injury prior to discharge, chronic lung disease at 36 weeks' gestational age, proven necrotising enterocolitis prior to discharge, retinopathy of prematurity (any stage) prior to discharge, and intraventricular haemorrhage grade III or IV prior to discharge.

All trials planned to report severe adverse reactions, but only Hansen 2023 observed their occurrence ("Four serious adverse reactions, all severe skin injuries, occurred in the NIRS group").

Only two trials reported retinopathy of prematurity (stage ≥ 3) prior to discharge (Hansen 2023; Hyttel-Sorensen 2015), only Hyttel-Sorensen 2015 reported major neurodevelopmental disability in children aged 18 to 24 months, and only Pichler 2023 reported abnormal fidgety movements.

Excluded studies

We excluded one study during the full-text screen, as it did not measure relevant outcomes. For details, see [Supplementary material 3](#).

Ongoing studies

We identified five ongoing RCTs (ACTRN12621000778886 [102]; ChiCTR2100051928 [103]; NCT04439968 [104]; NCT06926946 [105]; Vestager 2023 [106, 107]). Two of these trials will include preterm infants born before 29 weeks' gestation with an estimated sample size of 100 infants (ACTRN12621000778886; NCT04439968). Two trials will include preterm infants born after 28 weeks' gestation, with estimated sample sizes of 3000 infants (Vestager 2023) and 1000 infants (NCT06926946). ChiCTR2100051928 will include preterm infants born before 37 weeks' gestation, with an estimated sample size of 300 infants. [Supplementary material 4](#) presents detailed information.

Risk of bias in included studies

[Supplementary material 2](#) presents all risk of bias judgements with justifications. [Figure 2](#) and [Figure 3](#) present the results visually.

Figure 2. Review authors' judgements about each risk of bias item presented as percentages across all included studies

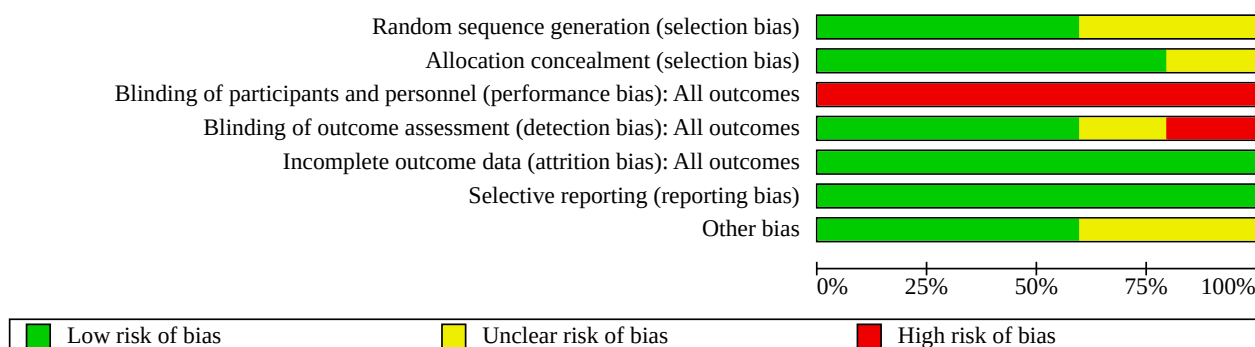


Figure 3. Review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Hansen 2023	+	+	-	+	+	+	?
Hyttel-Sorensen 2015	+	+	-	+	+	+	?
Pichler 2016	?	+	-	+	+	+	+
Pichler 2018	?	?	-	-	+	+	+
Pichler 2023	+	+	-	?	+	+	+

In three trials, we considered there was low risk of bias arising from random sequence generation and allocation concealment (Hansen 2023; Hyttel-Sorensen 2015; Pichler 2023). In two trials, selection bias was unclear (Pichler 2016; Pichler 2018).

Due to the nature of the intervention, we judged all trials at high risk of performance bias, because the group allocation was evident to the personnel and parents. Three trials had a low risk of detection bias, as all outcomes were assessed without knowledge of the allocated interventions (Hansen 2023; Hyttel-Sorensen 2015; Pichler 2016). We judged Pichler 2023 at unclear risk of detection bias and Pichler 2018 at high risk.

We did not identify any attrition or reporting bias in the studies analysed.

In three trials, we found no other source of bias that could influence the intervention effect (Pichler 2016; Pichler 2018; Pichler 2023). In the other two trials, the risk of bias from other potential sources was unclear.

Synthesis of results

Supplementary material 5 presents all analyses. Summary of findings 1 and Supplementary material 7 provide GRADE ratings and justifications.

Primary analysis

Critical outcomes

All-cause mortality

Prior to discharge

All five trials reported this outcome. Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in all-cause mortality prior to discharge (RR 0.99, 95% CI 0.83 to 1.19; $I^2 = 41\%$; 5 studies, 2415 participants; moderate-certainty evidence; Analysis 1.1). No events occurred in Pichler 2018. For Pichler 2016, we obtained outcome data from the trial authors.

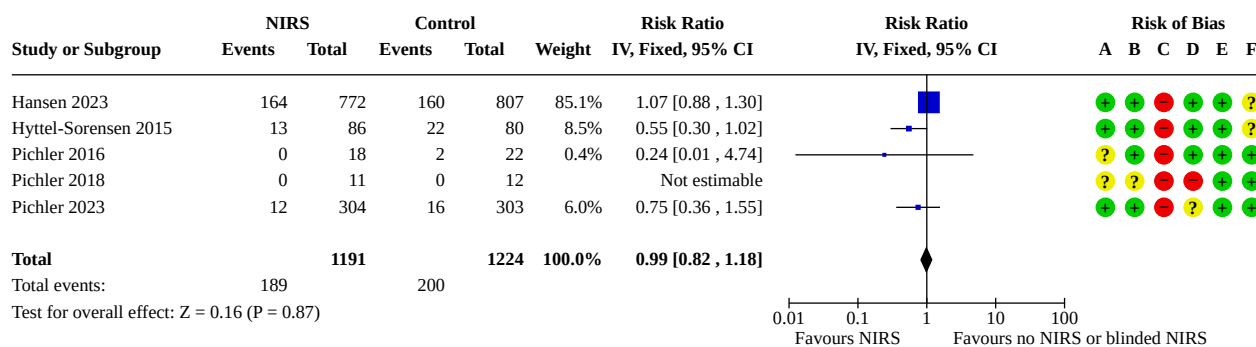
At two years of life

One trial reported this outcome (Hyttel-Sorensen 2015). The evidence is very uncertain about the effect of continuous cerebral NIRS monitoring compared to blinded NIRS monitoring on all-cause mortality at two years of corrected age (RR 0.55, 95% CI 0.30 to 1.02; 1 study, 166 participants; very low-certainty evidence; Analysis 1.2).

At longest follow-up

All five trials reported this outcome. The longest follow-up was at discharge for four trials (Hansen 2023; Pichler 2016; Pichler 2018; Pichler 2023), and at two years of age for one trial (Hyttel-Sorensen 2015). Continuous cerebral NIRS monitoring likely results in little to no difference in all-cause mortality at longest follow-up (RR 0.99, 95% CI 0.82 to 1.18; $I^2 = 46\%$; 5 studies, 2415 participants; moderate-certainty evidence; Figure 4; Analysis 1.3). No events occurred in Pichler 2018. For Pichler 2016, we obtained outcome data from the trial authors.

Figure 4. All cause mortality at longest follow-up



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Major neurodevelopmental disability in children aged 18 to 24 months

One trial reported this outcome (Hyttel-Sorensen 2015). The evidence is very uncertain about the effect of continuous cerebral NIRS monitoring compared to blinded NIRS monitoring on major neurodevelopmental disability at 18 months' chronological age

(RR 1.28, 95% CI 0.50 to 3.29; 1 study, 115 participants; very low-certainty evidence; Analysis 1.4).

Major brain injury prior to discharge

All five trials reported this outcome. Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in major brain injury diagnosed by brain ultrasound prior to discharge (RR 0.99, 95% CI 0.84 to 1.17; $I^2 = 13\%$; 5 studies, 2415 participants; moderate-certainty evidence; Analysis 1.5). For two trials, we obtained outcome data from trial authors (Pichler 2016; Pichler 2018).

Important outcomes**Intraventricular haemorrhage grade III or IV prior to discharge**

All five trials reported this outcome. Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in intraventricular haemorrhage grade III or IV prior to discharge (RR 0.93, 95% CI 0.76 to 1.13; $I^2 = 0\%$; 5 studies, 2415 participants; moderate-certainty evidence; Analysis 1.6). For two trials, we obtained outcome data from trial authors (Pichler 2016; Pichler 2018).

Cystic periventricular leukomalacia prior to discharge

Four trials reported this outcome (Hansen 2023; Pichler 2016; Pichler 2018; Pichler 2023). Continuous cerebral NIRS monitoring

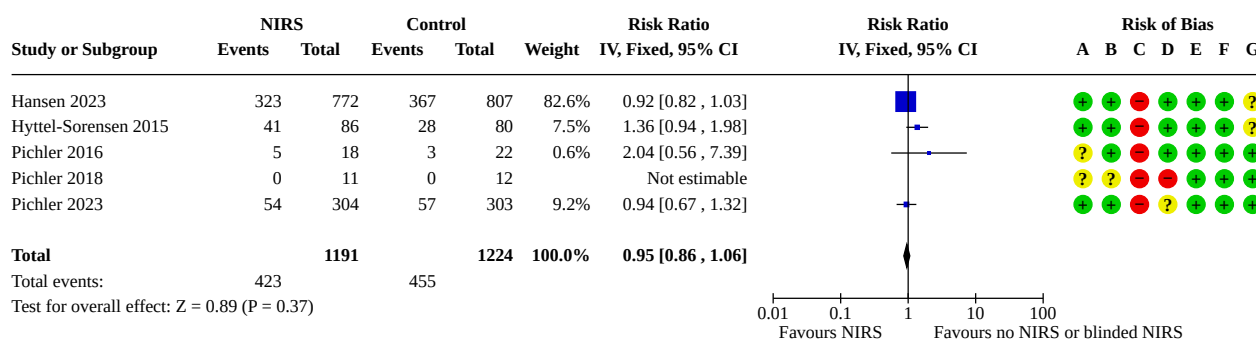
compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in cystic periventricular leukomalacia prior to discharge (RR 1.25, 95% CI 0.81 to 1.91; $I^2 = 30\%$; 4 studies, 2249 participants; moderate-certainty evidence; Analysis 1.7). No events occurred in two trials (Pichler 2016; Pichler 2018). For two trials, we obtained outcome data from trial authors (Pichler 2016; Pichler 2018).

Air leak (any and gross, including pneumothorax or pneumomediastinum) prior to discharge

No trials evaluated air leak prior to hospital discharge.

Chronic lung disease (oxygen supplementation) at 36 weeks of gestational age

All five trials reported this outcome. Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in chronic lung disease at 36 weeks of gestational age (RR 0.95, 95% CI 0.86 to 1.06; $I^2 = 43\%$; 5 studies, 2415 participants; moderate-certainty evidence; Figure 5; Analysis 1.8). No events occurred in Pichler 2018. For two trials, we obtained outcome data from trial authors (Pichler 2016; Pichler 2018).

Figure 5. Chronic lung disease at 36 weeks of gestational age**Risk of bias legend**

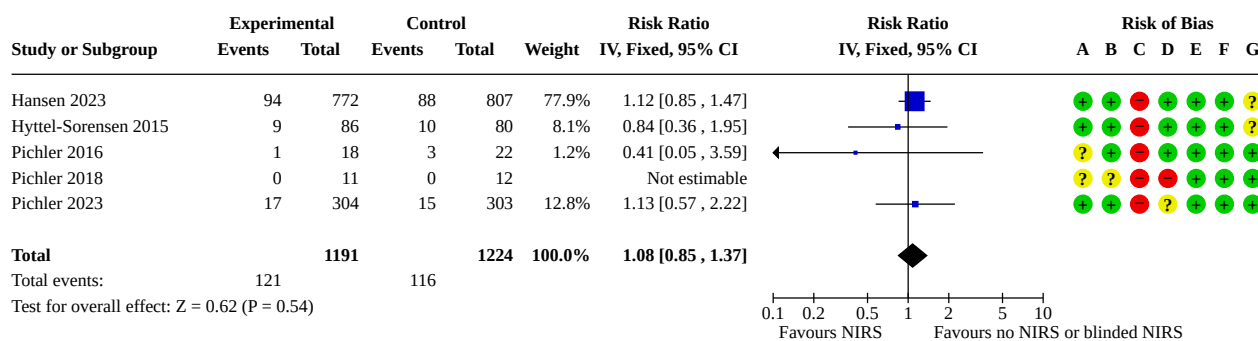
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias): All outcomes
- (F) Selective reporting (reporting bias)
- (G) Other bias

Proven necrotising enterocolitis prior to discharge

All five trials reported this outcome. Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in proven necrotising enterocolitis

prior to discharge (RR 1.08, 95% CI 0.85 to 1.37; $I^2 = 0\%$; 5 studies, 2415 participants; moderate-certainty evidence; Figure 6; Analysis 1.9). No events occurred in one Pichler 2018. For two trials, we obtained outcome data from trial authors (Pichler 2016; Pichler 2018).

Figure 6. Proven necrotising enterocolitis prior to discharge

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias): All outcomes
- (F) Selective reporting (reporting bias)
- (G) Other bias

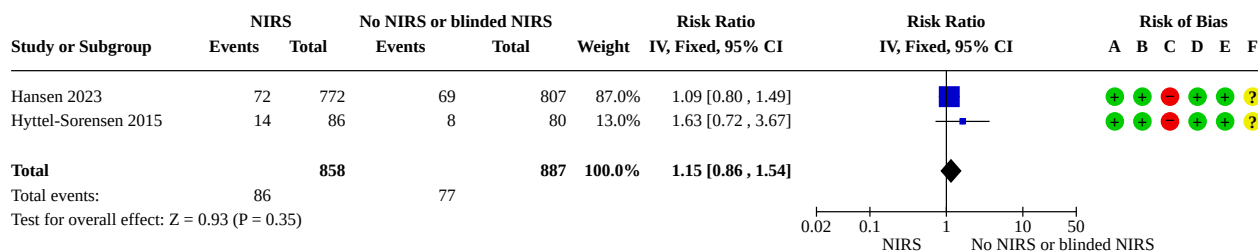
Retinopathy of prematurity (any stage) prior to discharge

All five trials reported this outcome. Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in retinopathy of prematurity of any stage prior to discharge (RR 1.15, 95% CI 0.90 to 1.47; I² = 0%; 5 studies, 2415 participants; moderate-certainty evidence; Analysis 1.10). For two trials, we obtained outcome data from trial authors (Pichler 2016; Pichler 2018).

Retinopathy of prematurity (stage ≥ 3) prior to discharge

Two trials reported this outcome (Hansen 2023; Hyttel-Sorensen 2015). Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in retinopathy of prematurity (stage ≥ 3) prior to discharge (RR 1.15, 95% CI 0.86 to 1.54; I² = 0%; 2 studies, 1745 participants; moderate-certainty evidence; Figure 7; Analysis 1.11).

Figure 7. Retinopathy of prematurity (stage ≥ 3) prior to discharge

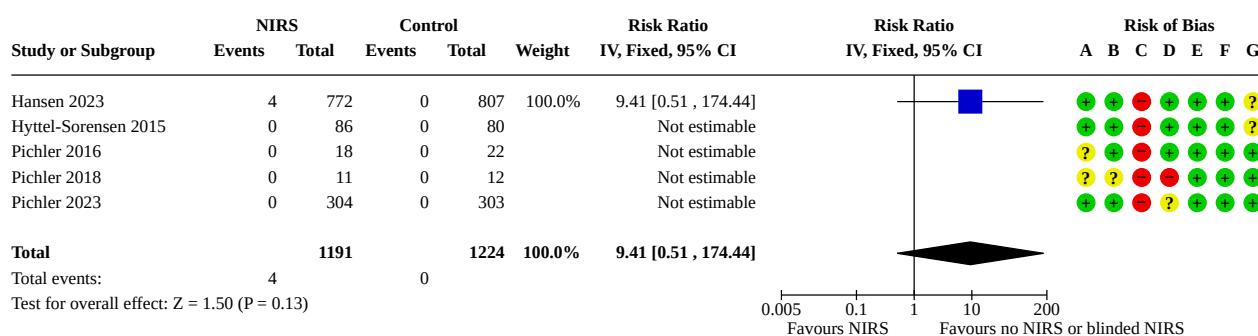
**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Severe adverse reactions prior to discharge

All five trials reported this outcome. Continuous cerebral NIRS monitoring compared to no NIRS or blinded NIRS monitoring likely results in little to no difference in severe adverse reactions prior

to discharge (RR 9.41, 95% CI 0.51 to 174.44; I² not applicable; 5 studies, 2415 participants; moderate-certainty evidence; Figure 8; Analysis 1.12). No events occurred in four trials (Hyttel-Sorensen 2015; Pichler 2018; Pichler 2023; Pichler 2016).

Figure 8. Severe adverse reactions prior to discharge**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias): All outcomes
- (F) Selective reporting (reporting bias)
- (G) Other bias

Abnormal fidgety movements

Pichler 2023 reported fidgety movements (abnormal at six to 20 weeks' corrected age). Continuous cerebral NIRS monitoring compared to blinded NIRS monitoring may result in little to no difference in fidgety movements (RR 0.87, 95% CI 0.24 to 3.12; 1 study, 171 participants; low-certainty evidence; Analysis 1.13).

Trial Sequential Analysis

The TSA results (11 figures) are available at <https://doi.org/10.6084/m9.figshare.29948006.v1>. **Supplementary material 8** compares the downgrading for precision with GRADE versus TSA.

All-cause mortality**Prior to discharge**

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 0.87, 95% CI 0.23 to 3.36). This TSA implies downgrading the evidence certainty by three levels for imprecision.

At two years of life

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 0.55, 95% CI 0.05 to 5.88). This TSA implies downgrading the evidence certainty by three levels for imprecision.

At longest follow-up

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 0.84, 0.19 to 3.68). This TSA implies downgrading the evidence certainty by three levels for imprecision.

Major neurodevelopmental disability in children aged 18 to 24 months

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR. There was too little information

to calculate boundaries of the alpha spending adjusted CI. This TSA implies downgrading the evidence certainty by three levels for imprecision.

Major brain injury prior to discharge

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 0.96, 95% CI 0.55 to 1.66). This TSA implies downgrading the evidence certainty by three levels for imprecision.

Important outcomes**Intraventricular haemorrhage grade III or IV prior to discharge**

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 0.93, 95% CI 0.66 to 1.30). This TSA implies downgrading the evidence certainty by two levels for imprecision.

Cystic periventricular leukomalacia prior to discharge

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR. There was too little information to calculate boundaries of the alpha spending adjusted CI. This TSA implies downgrading the evidence certainty by three levels for imprecision.

Chronic lung disease (oxygen supplementation) at 36 weeks of gestational age

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 1.03, 95% CI 0.68 to 1.57). This TSA implies downgrading the evidence certainty by two levels for imprecision.

Proven necrotising enterocolitis prior to discharge

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 1.08, 95% CI 0.64 to 1.82). This TSA implies downgrading the evidence certainty by three levels for imprecision.

Retinopathy of prematurity (stage ≥ 3) prior to discharge

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 1.15, 95% CI 0.39 to 3.43). This TSA implies downgrading the evidence certainty by three levels for imprecision.

Severe adverse reactions prior to discharge

We did not perform a TSA because there were no events.

Abnormal fidgety movements

According to the TSA result, there is no firm evidence for an intervention effect of 20% RRR (RR 0.87, 95% CI 0.24 to 3.08). This TSA implies downgrading the evidence certainty by three levels for imprecision.

Equity assessment

We did not examine any equity-related characteristics in this review. However, the included studies were conducted mainly in high-income countries, given the cost of NIRS technology.

Reporting biases

We provide our assessment of the risk of selective outcome reporting for each included study in [Risk of bias in included studies](#).

Because we included only five trials, we were unable to explore publication bias using funnel plots.

DISCUSSION

Summary of main results

We evaluated the benefits of continuous NIRS monitoring in preventing death or neurodevelopmental disability in very preterm infants. We included five trials with 2415 preterm infants, comparing NIRS monitoring with either no NIRS or blinded NIRS monitoring.

NIRS monitoring likely results in little to no difference in all-cause mortality at longest follow-up and in major brain injury diagnosed by brain ultrasound prior to discharge. The evidence is very uncertain about the effect of NIRS monitoring on major neurodevelopmental disability in children aged 18 to 24 months. NIRS monitoring likely results in little to no difference in chronic lung disease at 36 weeks' gestational age, proven necrotising enterocolitis prior to discharge, retinopathy of prematurity (stage ≥ 3) prior to discharge, and severe adverse reactions prior to discharge.

The evidence is very uncertain about the effect of NIRS monitoring on all-cause mortality at two years of life. Moreover, NIRS monitoring likely results in little to no difference in all-cause mortality prior to discharge, intraventricular haemorrhage grade III or IV prior to discharge, and cystic periventricular leukomalacia prior to discharge.

We identified five ongoing trials, one of which is aiming to enrol 3000 newborn infants by February 2029 (Vestager 2023).

Limitations of the evidence included in the review

The certainty of the evidence was moderate for six of the seven outcomes listed in [Summary of findings 1](#). We downgraded these

outcomes for serious imprecision due to wide CIs. For major neurodevelopmental disability in children aged 18 to 24 months, we downgraded the evidence certainty by three levels for extremely serious imprecision due to very wide CIs, small sample size, and few events. We were unable to investigate publication bias with funnel plots because the review included only five trials. Therefore, although we did not downgrade the certainty of evidence for publication bias, we cannot rule out a potential impact. Similarly, the paucity of trials precluded subgroup analysis. [Supplementary material 7](#) presents the GRADE ratings for the outcomes not included in the summary of findings table. These outcomes were downgraded by one, two, or three levels for imprecision.

We also compared the downgrading of imprecision according to GRADE criteria and according to our TSA, finding that TSA led to more frequent downgrading by three levels to very low certainty ([Supplementary material 8](#)). This observation aligns with multiple publications [108, 109]. TSA downgrading seems quite robust to our choices of parameters, as more restrictive choices would not lead to further downgrading. The downgrading of imprecision with TSA depends on our choice of parameters for calculating the DARIS. Had we used, for example, smaller alpha values to keep our family-wise error rate at 5% or below, and had we used more plausible intervention effects (e.g. a 2% RRR), we would have seen much wider TSA-adjusted CIs, but this would not have affected our GRADE assessments.

The studies included a variety of participants with different start times and duration of NIRS monitoring, factors that could lead to clinical and statistical heterogeneity. The studies mostly took place in high-income countries (the only exception was Hansen 2023, where one of 17 centres was in a low-income country), possibly owing to the high cost of NIRS technology.

Since combining NIRS monitoring with a treatment guideline constitutes a complex intervention, variations in protocols used across NICUs to interpret and manage infants based on their cerebral NIRS values might directly influence how the intervention impacts all clinically relevant outcomes. In some trials, participating sites used different NIRS devices. However, randomisation at the site level is likely to have minimised potential biases arising from this variability.

Limitations of the review processes

For this update, we revised the review methodology according to the latest Cochrane guidance [54]. These changes included refining our outcomes to focus on those most relevant to infants, their families, and guideline developers. We made all methodological changes prior to screening studies for inclusion and in agreement with Cochrane Neonatal, following MECIR standard U1. Detailed information and rationale are provided in the [Methods](#) section.

To assess clinical and statistical heterogeneity, we considered exploring the effect of varying intervention durations and gestational age through subgroup analyses. However, there were too few eligible trials to conduct these analyses for any outcome (fewer than five sufficiently informative trials per comparison). Such subgroup analyses could have provided a clearer understanding of the influence of these potential confounders.

We decided to include only randomised and quasi-randomised studies. As a result, our review was bound to focus most on benefits, as randomised trials tend to provide less robust reporting of harms compared with observational studies, which usually have larger sample sizes [110]. If we find clinically relevant benefits of NIRS in future updates of this review, we might need to conduct a systematic review on harms, including data from observational studies.

Agreements and disagreements with other studies or reviews

We are aware of only one systematic review on the use of cerebral NIRS in very preterm infants: the previous version of this review, which included one study with 166 infants [52]. We included four additional trials in this update, resulting in a total sample size of 2415 infants. This substantial increase in the statistical power resulted in moderate certainty of evidence for most outcomes, thus providing a more useful evidence base for clinicians and guideline developers. One previous systematic review on cerebral NIRS in both children and adults included 25 trials, but only two in the neonatal population [111]. Narrative reviews have suggested practical guidelines on the management of abnormal cerebral NIRS patterns and highlighted the need for international consensus [112]. Moreover, researchers have indicated that the effectiveness of cerebral NIRS needs a more solid evidence base before it can be recommended as standard practice in the management of very preterm infants [51].

AUTHORS' CONCLUSIONS

Implications for practice

Compared with conventional monitoring, near-infrared spectroscopy (NIRS) monitoring in very preterm infants likely results in little to no difference in all-cause mortality at longest follow-up and in major brain injury diagnosed by brain ultrasound prior to discharge. We are unsure about its effect on major neurodevelopmental disability in children aged 18 to 24 months and on mortality at two years of life. NIRS monitoring likely results in little to no difference in chronic lung disease at 36 weeks' gestational age, proven necrotising enterocolitis prior to discharge, severe retinopathy of prematurity prior to discharge, and severe adverse reactions prior to discharge.

Our review found no evidence that NIRS monitoring provides clinical benefits to very preterm infants, while data on potential harms are very limited.

Equity-related implications for practice

The included studies were conducted predominately in high-income countries, probably owing to the high cost of NIRS technology. Clinicians and policymakers in low-income settings should critically evaluate whether NIRS monitoring represents a worthwhile investment, as it has no proven clinical benefits for preterm infants.

Implications for research

The performance characteristics of available NIRS devices require more rigorous comparative evaluation. Furthermore, future randomised clinical trials in very preterm infants should implement NIRS monitoring from birth through the early cardiorespiratory

stabilisation period to more accurately determine its potential benefits. Long-term neurodevelopmental outcomes (e.g. Bayley scores at two years) remain the most clinically meaningful endpoints, but detecting a 10% relative risk reduction with an alpha of 0.05 and a beta of 0.20 would require sample sizes of approximately 7104 infants for a composite outcome of death or major cerebral injury, raising major feasibility concerns. Surrogate outcomes, including electrophysiological markers of hypoxic brain injury, may offer a more practicable alternative for early-phase trials, but they must be rigorously validated against clinical outcomes.

Equity-related implications for research

Current research on cerebral NIRS is largely limited to high-income countries, creating a significant equity gap in evidence generation. To promote more inclusive and globally relevant neonatal care, future studies should prioritise settings in low- and middle-income countries (LMICs), assess the feasibility and cost-effectiveness of NIRS in resource-constrained environments, and explore scalable alternatives. Collaborative research with LMIC partners is essential to ensure that evidence reflects diverse health system realities and supports equitable implementation across varying contexts.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD011506.pub3](https://doi.org/10.1002/14651858.CD011506.pub3).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 GRADE assessment for outcomes not included in the summary of findings table

Supplementary material 8 Comparasion of downgrading levels of precision in the certainty of the evidence using GRADE versus Trial Sequential Analysis (TSA)

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Prakesh S Shah, Professor, Department of Pediatrics, Mount Sinai Hospital and University of Toronto, Toronto, Canada.
- Managing Editor (provided editorial guidance to authors, edited the article): Justin Mann, Cochrane Central Editorial Service.
- Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer-reviewer comments and supported editorial team): Cynthia Stafford, Cochrane Central Editorial Service.
- Copy Editor (copy editing and production): Julia Turner, Cochrane Central Production Service
- Peer-reviewers (provided comments and recommended an editorial decision): Yun Yu, Department of Anesthesiology, Beijing Tiantan Hospital, Capital Medical University, Beijing, China (clinical/content review); Sowndharya PK, Doctor of Pharmacy, Amrita Vishwa Vidyapeetham (patient and public review); Clare Miles, Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review). One additional peer reviewer provided clinical/content peer review but chose not to be publicly acknowledged.

Contributions of authors

Two authors involved in previous published versions of this review in 2017 are no longer included on the author byline: Hyttel-Sorensen S and Als-Nielsen B. Some of the content retained in this review reflects their contributions.

MF developed the search strategies and conducted the literature search.

EB, MB, and MCI screened titles and abstracts and assessed full-text articles to identify relevant studies.

EB and MCI independently extracted data and assessed the risk of bias in included studies. MB resolved any disagreements.

EB, MB, and MCI transferred study references from Covidence to RevMan.

MB led a discussion involving all review authors to assess the certainty of the evidence, whereas four review authors (CG, GG, MB, MCI) applied the Trial Sequence Analysis in a process led by CG.

When additional data for included and ongoing studies were required, EB and MB contacted study investigators/authors by email for clarification.

All review authors contributed to writing the text of the review.

All review authors reviewed, edited, and approved the final version of the review.

Declarations of interest

EB has no known conflicts of interest.

MCI has no known conflicts of interest.

MF is the Managing Editor and Information Specialist of Cochrane Neonatal; she did not participate in the editorial acceptance of this review.

What's new

GG has done research using near-infrared spectroscopy for 20 years and therefore could be perceived to be biased by academic interests. He participated in the SafeBoosC trial (Hyttel-Sorensen 2015), which could bias his views.

CG participated in the SafeBoosC trial (Hyttel-Sorensen 2015), which could bias his views.

MB is an Associate Editor for the Cochrane Neonatal Group. However, he had no involvement in the editorial processing of this review.

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MB is employed by this organisation

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Registration and protocol

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Original review (2017): doi.org/10.1002/14651858.CD011506.pub2

Data, code and other materials

The following are made available for download for users of the Cochrane Library: full search strategies for each database; full citations of each unique report for all studies included, ongoing or waiting classification, or excluded at the full text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions are obtained for such use. Analyses and data management are conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Covidence and [Supplementary material 6](#) are available from the authors on reasonable request.

Date	Event	Description
7 January 2026	New citation required and conclusions have changed	We updated searches in 2025 and found four new eligible studies for inclusion.
7 January 2026	New search has been performed	Search strategies revised and rerun without date limits.

History

Protocol first published: Issue 2, 2015

Review first published: Issue 9, 2017

Date	Event	Description
3 October 2017	Amended	Minor edits have been made for clarity in the Plain Language Summary.

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ADDITIONAL TABLES

Table 1. Overview of the included studies

Study ID	Country	GA of infants	Sample size (NIRS/control)	Comparison	Initiation of NIRS monitoring	Duration of NIRS monitoring	Outcomes of Summary of findings 1 with available data
Hansen 2023	17 countries in Asia, Europe, and North America	< 28 weeks	1601 (785/816)	NIRS monitoring versus no NIRS monitoring	In first 6 hours after birth	Planned for 72 hours	- All-cause mortality at longest follow-up - Major brain injury prior to discharge - Chronic lung disease at 36 weeks' GA - Proven necrotising enterocolitis prior to discharge - Retinopathy of prematurity (stage ≥ 3) prior to discharge - Severe adverse reactions prior to discharge
Hytel-Sorensen 2015	France, Denmark, Spain, Ireland, the Netherlands, Austria, Italy, UK	< 28 weeks	166 (86/80)	NIRS monitoring versus blinded NIRS monitoring	In first 3 hours after birth	Mean of 67.9 hours	- All-cause mortality at longest follow-up - Major neurodevelopmental disability in children aged 18–24 months - Major brain injury prior to discharge - Chronic lung disease at 36 weeks' GA - Proven necrotising enterocolitis prior to discharge - Retinopathy of prematurity (stage ≥ 3) prior to discharge - Severe adverse reactions prior to discharge
Pichler 2016	Austria and Canada	< 34 weeks	60 (30/30); only 40 infants with GA < 32 weeks	NIRS monitoring versus no NIRS monitoring	In first 2–3 minutes after birth	15 minutes	- All-cause mortality at longest follow-up - Major brain injury prior to discharge - Chronic lung disease at 36 weeks' GA - Proven necrotising enterocolitis prior to discharge - Severe adverse reactions prior to discharge
Pichler 2018	Austria	< 37 weeks	(98 (49/49); only 23 infants with GA < 32 weeks	NIRS monitoring versus blinded NIRS monitoring	In first 6 hours after birth	Mean of 24.4 hours	- All-cause mortality at longest follow-up - Major brain injury prior to discharge - Chronic lung disease at 36 weeks' GA - Proven necrotising enterocolitis prior to discharge - Severe adverse reactions prior to discharge

Table 1. Overview of the included studies *(Continued)*

Pichler 2023	6 countries in Europe and Canada	< 32 weeks	655 (329/326)	NIRS monitoring versus blinded NIRS monitoring	In first 2–3 minutes after birth	15 minutes	• All-cause mortality at longest follow-up • Major brain injury prior to discharge • Chronic lung disease at 36 weeks' GA • Proven necrotising enterocolitis prior to discharge • Severe adverse reactions prior to discharge
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GA: gestational age; **NIRS:** near-infrared spectroscopy.